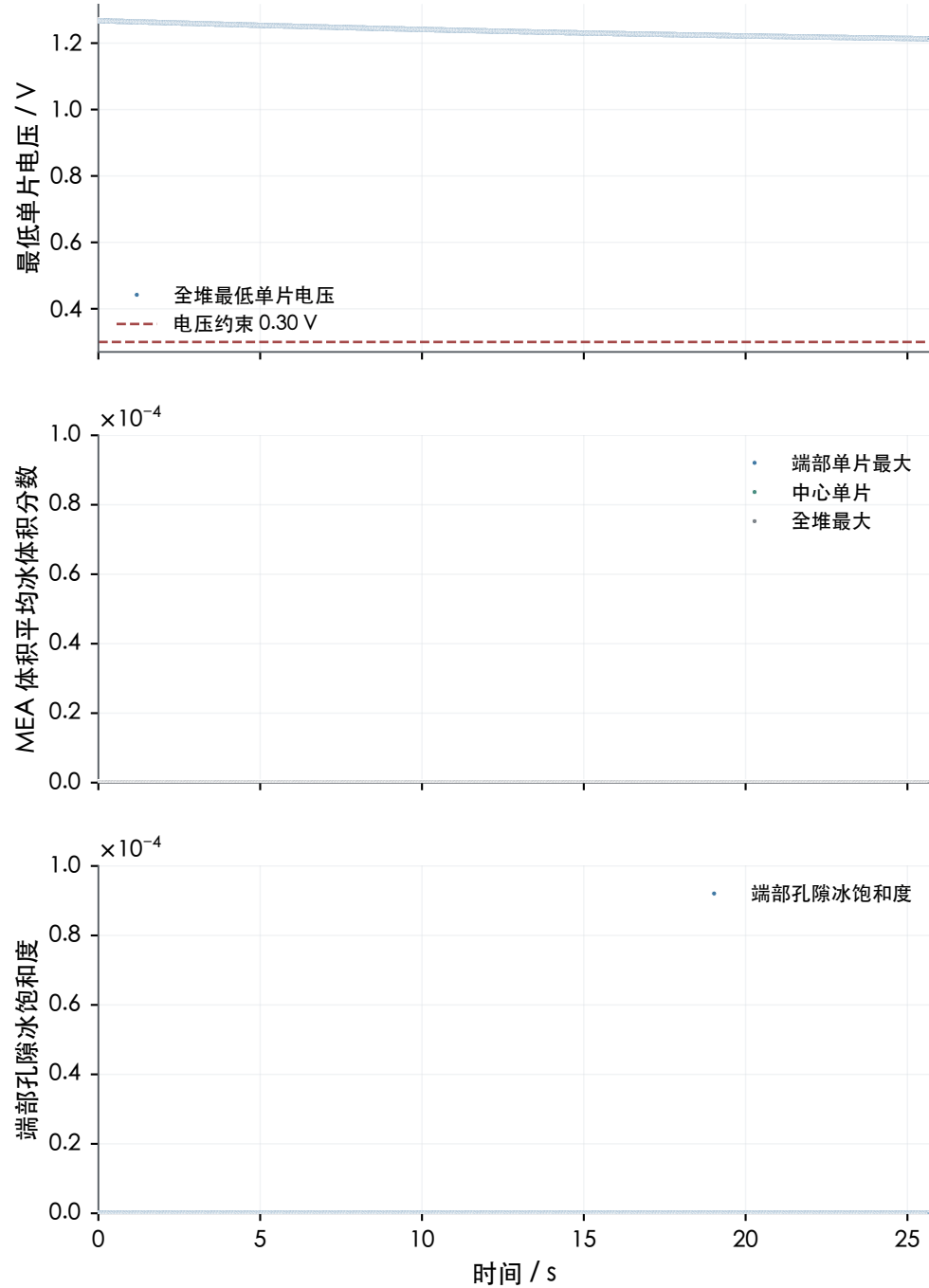
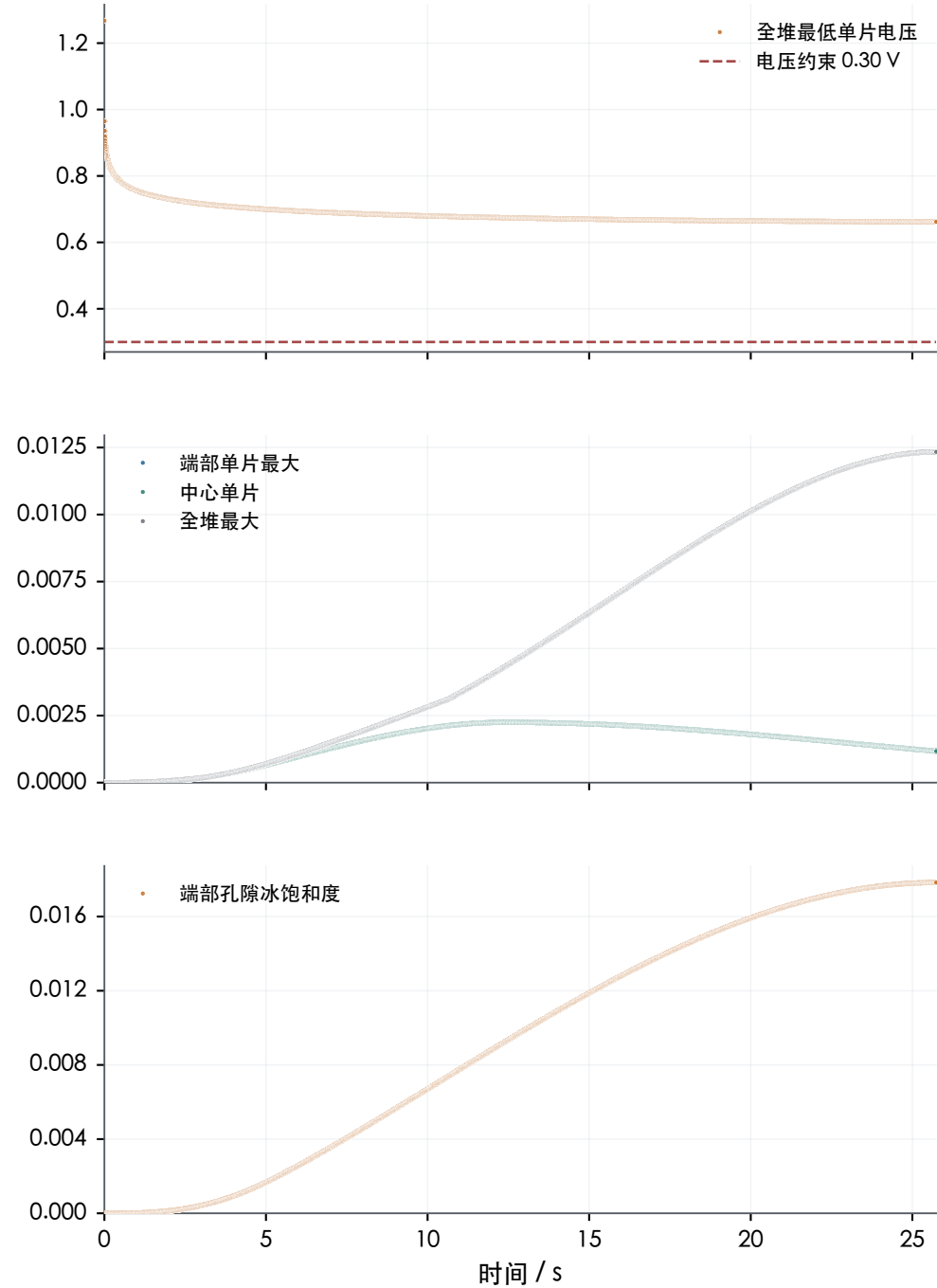


(a) P 纯预热



(b) C 协同加热



统计窗口与图01相同，未包含 P 关热后续加载。ice_bulk 含膜内冰、以整个 MEA 体积平均；孔隙冰饱和度仅描述多孔层局部孔隙占据，二者分母不同。体积分启动约束为 < 0.99 。